

AD-502-Machine Learning

Lecture 1 : Introduction to Machine Learning

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Course Code	AD502
Course Title	Machine Learning
Number of Credits	(3-0-2) [L-T-P]
Evaluation	70 [End Sem]+20 [Mid sem] + 10 [Assignment/Quiz] =100
Course Repository	https://github.com/rajesh101191/Machine-Learning-PIEMR-2026-AD502

Course Objective

- Understand the fundamental concepts, principles, and applications of Machine Learning.
- Explore the complete Machine Learning lifecycle, including data collection, preprocessing, model development, evaluation, and deployment.
- Apply supervised, unsupervised, and ensemble learning algorithms to solve real-world problems.
- Develop practical skills in implementing machine learning models using Python and Scikit-learn.
- Evaluate, compare, and interpret machine learning models using appropriate performance metrics and best practices.

COURSE LEARNING WORKFLOW

A Flipped Learning Approach for Better Understanding



WHY THIS WORKFLOW?



PREPARE

Build a strong foundation before coming to class.



ENGAGE

Actively participate in discussions and activities.



PRACTICE

Hands-on practice strengthens your concepts.



APPLY

Assignments help you apply knowledge to real problems.



IMPROVE

Feedback helps you track progress and achieve excellence.

★ Your consistent preparation, active participation and regular practice are the keys to your success in this course!

Content is referred from

- Mitchell, T. M. (1997). *Machine Learning*. New York: McGraw-Hill. ISBN: 0-07-042807-7
- Ng, A. Y. (2009). *CS229: Machine Learning*. Stanford University, Department of Computer Science.
- Ravindran, B. (2026). *Introduction to Machine Learning*. NPTEL Online Certification Course, Department of Computer Science and Engineering, Indian Institute of Technology Madras.
- Nwanganga, F., *Machine Learning with Python: Foundations*, LinkedIn Learning.
- Williams, K., *Artificial Intelligence Foundations: Machine Learning*, LinkedIn Learning.
- Anaconda Inc., *Machine Learning with Python Professional Certificate*, LinkedIn Learning.

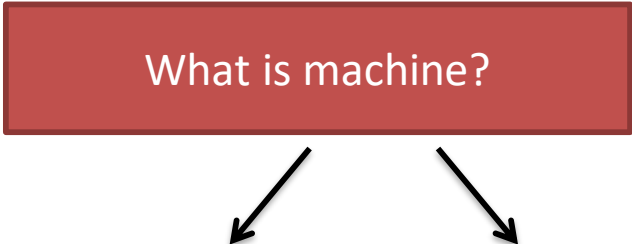
What is machine?

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A machine is a system that takes some input, performs a process, and produces a useful output.

What is machine?



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graph TD; A[What is machine?] --> B[A machine is a system that takes some input, performs a process, and produces a useful output.]; A --> C[A computer is also a machine. It takes data as input, processes it using instructions, and produces information as output.]
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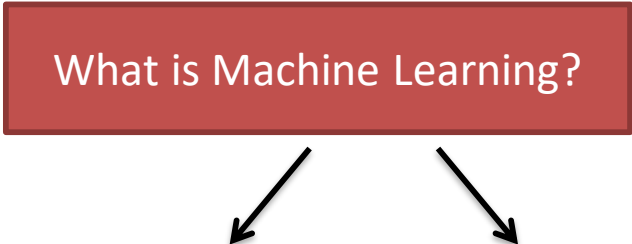
What is Learning?

What is Machine Learning?



Arthur Samuel (1995), Field of study that gives computer the ability to learn without being explicitly programmed.

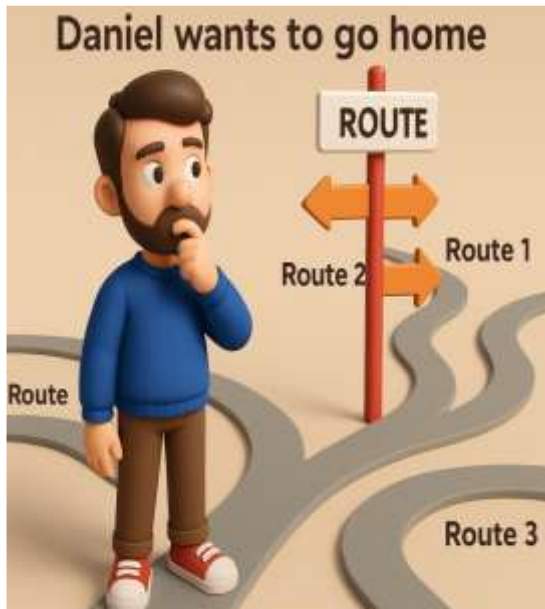
What is Machine Learning?

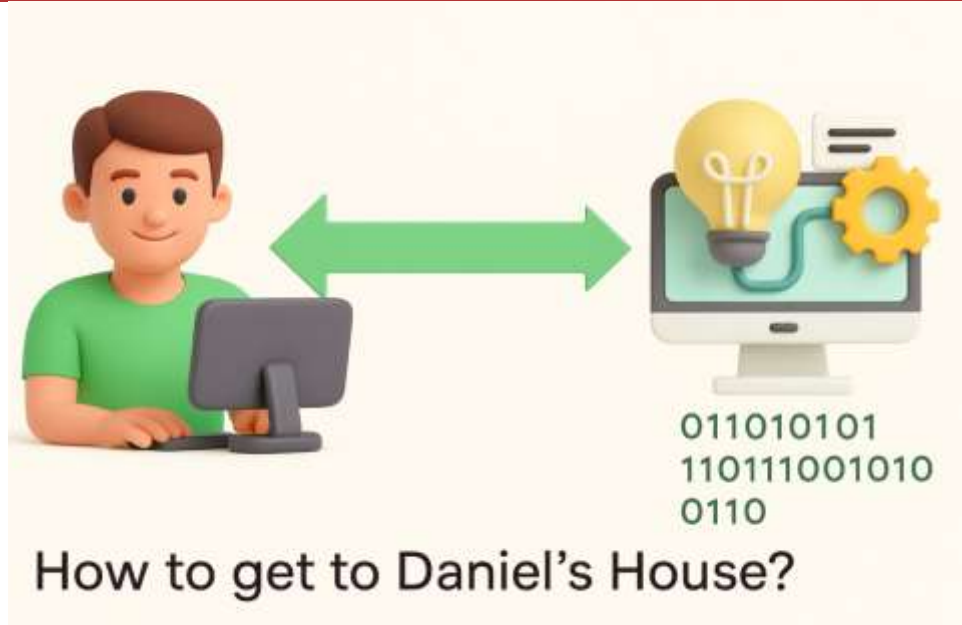


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If: Route 1 < Route 2

If: Route 1 < Route 3

If: Route 1 < Route 4

If: Route 1 < Route 5

Then: Pick Route 1

Introduction

WHEN TRADITIONAL PROGRAMMING WORKS VS. WHEN AI IS NEEDED

FINDING THE SHORTEST PATH



Can be programmed
using rules & logic

DETECTING EMOTIONS

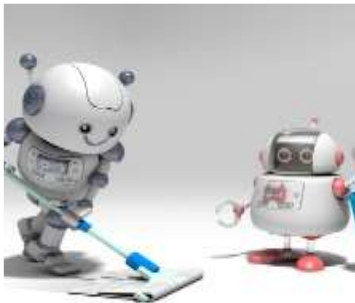


Needs MACHINE LEARNING
to learn patterns from data



- [Teachable Machine](#)
- <https://playground.tensorflow.org/>
- <https://quickdraw.withgoogle.com>

Types of Machine Learning



Supervised Machine Learning

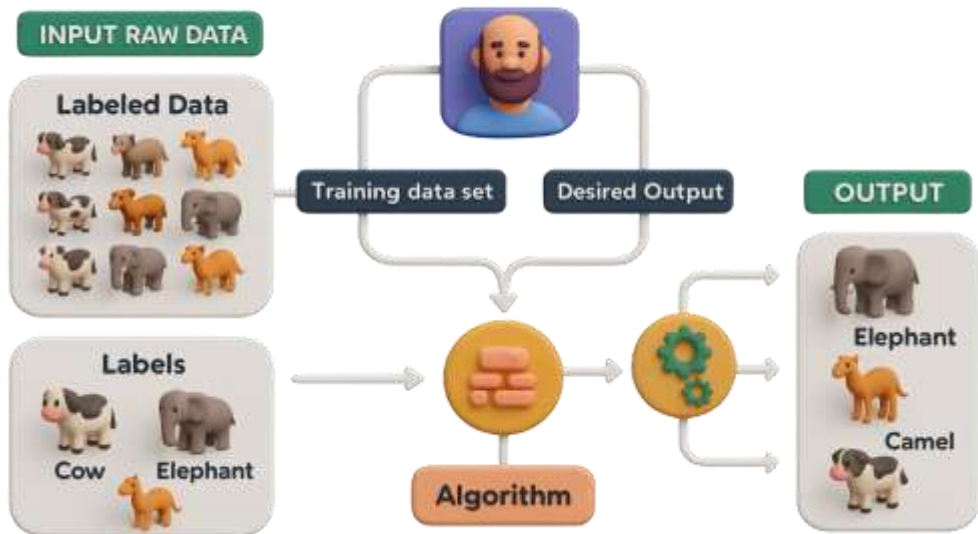


Unsupervised Machine Learning



Reinforcement Machine Learning

Types of Machine Learning: Supervised Learning



Classification

Independent Variable

Customer	Income	Loan Amount	Degree
James S.	\$30000	\$8000	No
Carol J.	\$28000	\$20000	No
Sarah P.	\$18000	\$15000	Yes
John J.	\$12000	\$21000	Yes
Saul B.	\$13000	\$35000	No
Alan Q.	\$13000	\$45000	Yes
Frank D.	\$15000	\$60000	No
Jane R.	\$26000	\$70000	Yes
Bill T.	\$28000	\$110000	Yes
Tom C.	\$16000	\$100000	No

Dependent Variable

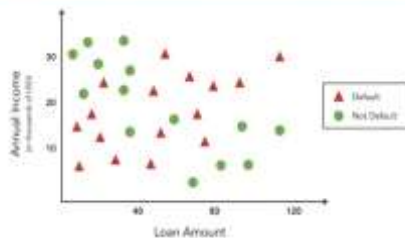
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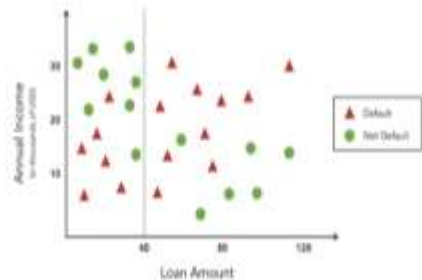
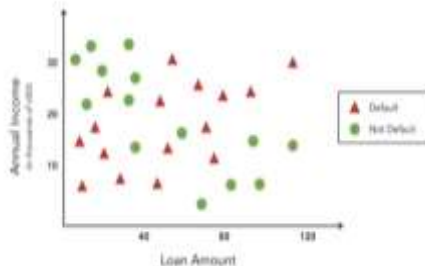
Dependent Variable

Recursive Partitioning



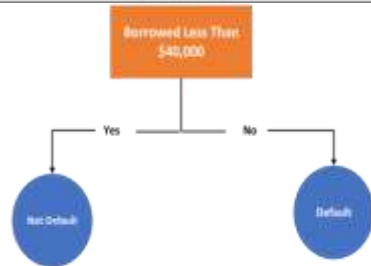
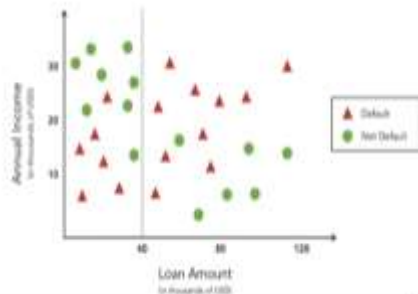
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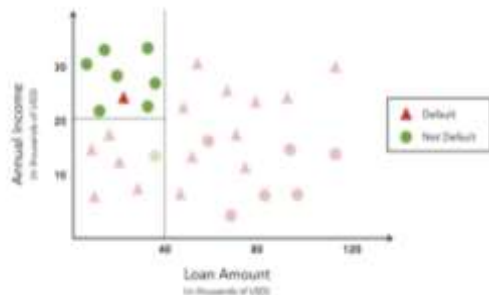
Types of Machine Learning: Supervised Learning

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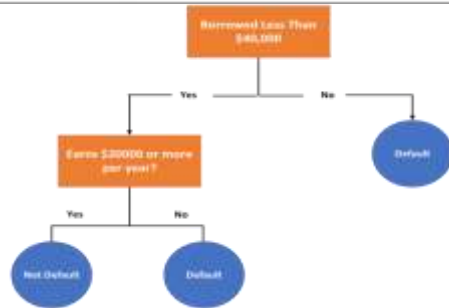
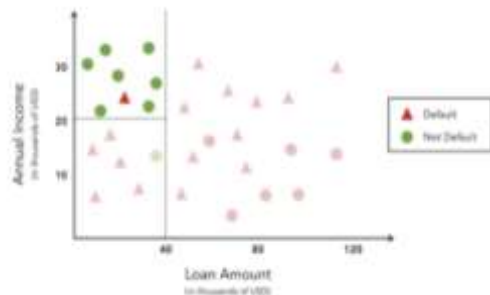
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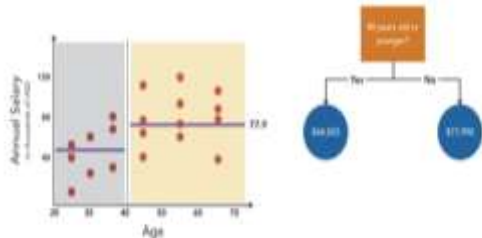
Types of Machine Learning: Supervised Learning

Regression

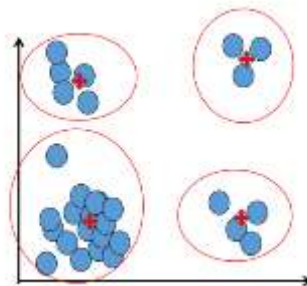
Independent Variables

Customer	Age	Education	Salary
James S.	25	Bachelor's	\$43,906
Carol J.	35	Doctorate	\$77,545
Sarah P.	55	Doctorate	\$95,675
John J.	25	Diploma	\$16,789
Saul B.	65	Master's	\$82,890
Alan Q.	55	Professional	\$118,000
Frank D.	45	Diploma	\$39,272
Jane R.	45	Bachelor's	\$62,472
Bill T.	45	Professional	\$115,455
Tom C.	35	Master's	\$61,686

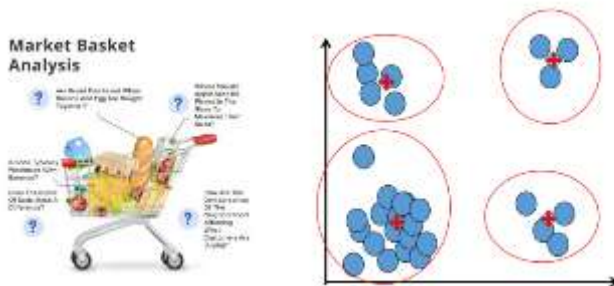
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Types of Machine Learning: Unsupervised Learning

- **Unsupervised learning:** The data have no target attribute.
- We want to explore the data to find some intrinsic structures in them.
- <https://ml-playground.com/#>



Market Basket Analysis

